

Details of Projects Undertaken in Professional Experience

Introduction

- **Name:** Ankit Sanjaybhai Talaviya
- **Companies:** Addicted Technologies (July 2023 - July 2024), Crest Infotech (July 2021 - June 2023)
- **Objective:** To provide an overview of the key projects undertaken, showcasing technical expertise and contributions.

Project Details

Addicted Technologies (July 2023 - July 2024):

Role: Frontend Developer, Team Leader

1. 3D-Emp (Desktop Software)

3D-Emp is a desktop application designed to provide an interactive house remodeling experience. The project involves loading detailed 3D models of houses created in Blender into the application using Three.js. Users can select specific walls of the house and apply different textures or colors, such as new tiles or paint, to visualize renovations in real time. The goal is to enhance user engagement by offering a realistic preview of post-renovation changes.

- **Technologies Used:** ReactJS, ElectronJS, ThreeJS
- **Tools:** Blender, Redux, Chrome Dev Tools, Git, Jira, Postman, Swagger
- **Key Contribution:**
 - **Project Initialization:**
 - Set up the project repository in GitLab and established the development environment, including configuration of ElectronJS and ReactJS for cross-platform functionality.
 - Created a modular project structure to ensure scalability and maintainability.
 - **Three.js Implementation:**
 - Loaded 3D house models exported from Blender into the application using Three.js loaders (e.g. GLTFLoader).
 - Applied camera controls (OrbitControls) to allow users to explore the house model in a 3D space.
 - **Interactive Wall Selection and Texture Application:**

- Implemented raycasting to detect and enable user interaction with specific wall meshes.
 - Developed logic to map user clicks to the appropriate mesh (wall) for accurate texture updates.
 - Integrated functionality to dynamically replace wall textures using `MeshStandardMaterial.map`, ensuring seamless real-time updates without performance degradation.
 - Provided customization options, such as tile patterns or paint colors, stored in a centralized Redux state for consistency and reusability.
- **Optimization and Debugging:**
- Optimized the rendering process by reducing the polygon count in Blender models and applying level-of-detail (LOD) techniques in Three.js.
 - Used Chrome Dev Tools to profile and fix performance bottlenecks, ensuring a smooth user experience.
- **Collaboration and Leadership:**
- Led a team of two developers, one designer, and one QA engineer, facilitating agile sprints and regular code reviews.
 - Coordinated with the designer to ensure accurate mapping of textures and alignment with user requirements.

2. FiMA – Finance Management (Desktop Software)

FiMA is a personal finance management desktop application developed using ReactJS and ElectronJS. It enables users to efficiently manage their day-to-day finances, providing features to track expenses and income, categorize transactions, and generate visual insights through graphs. The software also allows users to securely back up their financial data to Google Drive, ensuring data safety and accessibility.

- **Technologies Used:** ReactJS, ElectronJS
- **Tools:** Redux, Chrome Dev Tools, Git, Jira, Postman, Swagger
- **Key Contribution:**
 - **Design and Layout Development:**
 - Designed and implemented the base structure of the application, including a homepage, sidebar navigation, header, and footer, ensuring a consistent and user-friendly interface.
 - Used ReactJS components and styled them for responsiveness and cross-platform compatibility.

➤ **Expense and Income Management:**

- Built core functionalities for users to add, edit, and delete expenses and income entries, storing the data in a centralized Redux state.
- Implemented dynamic forms and validation to simplify data input for users.

➤ **Categorization of Transactions:**

- Developed features to categorize expenses by predefined categories such as Restaurant Bills, OTT Platform Subscriptions, Education, Health, and Fitness, making it easier for users to organize and analyze their finances.

➤ **Data Visualization:**

- Integrated libraries to generate interactive graphs and charts (bar charts and pie charts) for users to visually understand their financial trends.
- Ensured graphs updated dynamically based on user input and selected date ranges.

➤ **Data Backup to Google Drive:**

- Implemented functionality to allow users to securely back up their financial data to Google Drive using API integration, ensuring data recovery in case of application reinstallation.

Crest Infotech (Working Student – Frontend) (July 2021 – June 2023):

Role: Working Student / Frontend Developer

I joined Crest Infotech as a working student during my bachelor's degree, which provided me with my first hands-on experience in a professional software development environment. The role allowed me to gain valuable insights into the structured management of codebases, effective project workflows, and the dynamics of teamwork in a professional setting.

1. CrestPMS (Project Management Software)

CrestPMS is a project management software designed to enhance productivity tracking and workflow efficiency. It includes advanced features like automatic screenshot capture of employee activity, auto-logout functionality triggered after 7 minutes of inactivity, and a key and mouse tracking system to monitor engagement. The software is built using ElectronJS, ensuring cross-platform compatibility.

- **Technologies Used:** ElectronJS
- **Tools:** Chrome Dev Tools, Git, Jira, Redux, Postman, Swagger
- **Link:** www.crestpms.com
- **Key Contribution:**
 - **Bug Analysis and Debugging:**

- Investigated issues where the application could capture screenshots on Apple devices but was limited to a single monitor and failed to capture screens from an additional monitor connected via HDMI.
- Identified limitations in the ElectronJS desktopCapturer API and its configuration for handling multiple displays.
- Resolved issues where mouse and keyboard event tracking via ioHooks (<https://wilix-team.github.io/iohook/>) was not functioning properly.
- **Feature Redevelopment:**
 - Reconfigured the desktopCapturer to enable multi-monitor screenshot capabilities, ensuring that the application could seamlessly capture activity across all connected screens.
 - Debugged and updated the ioHook integration to reliably track mouse movements and keyboard events across different system configurations.
- **Testing and Optimization:**
 - Performed extensive testing on multiple hardware setups, including Linux, Mac M1, Mac Intel, and Windows systems with various monitor configurations, to ensure consistency in functionality.
 - Optimized the event tracking system to reduce latency and improve performance while capturing real-time user activities.
- **Collaboration and Deployment:**
 - Coordinated with the development team to integrate the updated features into the existing application.
 - Assisted in rebuilding the application with the latest ElectronJS version to ensure compatibility and stability across platforms.

2. Pawfect Admin Panel (Web Application)

The Pawfect Admin Panel is a web application built using ReactJS and Bootstrap, designed to provide administrators with an efficient interface for managing both customers and employees. The platform includes features for real-time communication, notifications, and seamless user interaction, offering a comprehensive solution for administrative tasks.

- **Technologies Used:** ReactJS, Bootstrap
- **Tools:** Chrome Dev Tools, Git, Jira, ioSocket, Redux, Postman, Swagger
- **Key Contributions:**
 - **Project Initialization and Design:**

- Set up the project structure from scratch, establishing the foundation for a scalable and maintainable codebase.
- Designed and implemented a responsive user interface using ReactJS and Bootstrap, ensuring a seamless experience across devices.
- **Real-Time Communication and Notifications:**
 - Integrated ioSocket to enable real-time chat functionality, allowing administrators to communicate instantly with employees and customers.
 - Developed a real-time notification system using ioSocket, ensuring timely updates for critical events and activities.
- **Enhanced User Experience:**
 - Implemented infinite scrolling for efficient data presentation, enabling administrators to browse large datasets without performance issues.
 - Designed user-friendly interfaces and optimized the application for faster load times and responsiveness.
- **Bug Resolution and Optimization:**
 - Debugged and resolved complex issues in the application, ensuring stability and reliability in production.
 - Collaborated with the team to optimize code performance and enhance the user experience.

3. Crestmeds Application (React Native Clone Application)

CrestMeds is a React Native application designed as a clone of the popular pharmacy platform Netmeds. The application provides users with a seamless interface to browse, search, and purchase pharmaceutical products. Its goal was to replicate the functionality and design of Netmeds while leveraging the cross-platform capabilities of React Native to ensure compatibility with both Android and iOS devices.

- **Technologies Used:** React Native
- **Tools:** React Navigation, Chrome Dev Tools, Git, Redux, Postman, Swagger
- **Link:** www.crestinfotech.com/netmeds-pharmacy-clone-app/
- **Key Contributions:**
 - **Project Initialization:**
 - Set up the project's initial structure, ensuring scalability and maintainability by adhering to best practices in React Native development.
 - Configured essential dependencies and tools for smooth app development and testing.

➤ **React Native Navigation:**

- Implemented React Navigation for managing app screens, enabling intuitive user transitions between various features such as product browsing, cart management, and user profile.
- Configured navigation stacks, tab navigations, and deep linking for a cohesive user experience.

➤ **UI/UX Design Implementation:**

- Designed and implemented responsive and visually appealing layouts, ensuring that the app was user-friendly and maintained consistency with Netmeds' design standards.
- Utilized React Native components and styling to create a polished and intuitive interface.\

4. Room Key Hotel Management System (Web Application)

The Room Key Hotel Management System is a ReactJS-based web application designed to manage the room booking processes of a hotel chain. The platform provides a centralized interface for administrators and branch managers to streamline room availability, booking, and management, ensuring efficient operation across multiple hotel branches.

- **Technologies Used:** ReactJS, Redux
- **Tools:** Chrome Dev Tools, Git, Jira, Redux, Postman, Swagger
- **Key Contributions:**

➤ **API Integration:**

- Integrated APIs for room data, enabling seamless synchronization of room availability, bookings, and related information between the backend and frontend.
- Ensured the APIs were efficiently utilized to handle dynamic updates and real-time changes in the booking system.

➤ **State Management Using Redux:**

- Implemented Redux for managing application state, ensuring data consistency and ease of scalability.
- Developed robust reducers, actions, and a centralized store to handle complex state updates across the application.

➤ **Role-Based Access Control:**

- Designed and implemented a hierarchical access control system such as admins can access to all branches and operations, while Branch Managers has limited

access to operations specific to their branch, ensuring data privacy and role-specific functionality.

- Maintained project integrity by enforcing proper data isolation based on user roles.

5. ProductY (Web Application)

ProductY® is a comprehensive, user-friendly, web-based product and catalog management solution. The platform is designed to empower teams to create, manage, and maintain product information interactively online. Its drag-and-drop interface enables users to seamlessly create professional catalogs, retail sites, and other product outputs.

- **Technologies Used:** ReactJS
- **Tools:** Redux, Chrome Dev Tools, Git, Jira, Swagger, Postman
- **Link:** <https://producty.com/>
- **App Link:** <https://app.producty.com/>
- **Key Contributions:**
 - **Bug Fixing and Optimization:**
 - Started as a bug solver, identifying and resolving critical issues to ensure smooth application functionality.
 - **Technology Migration:**
 - Contributed to the complete technology transition of the project from its previous tech stack to ReactJS, ensuring compatibility and improved performance.
 - **Feature Development:**
 - Designed and implemented a Vector Graphics Editor, enabling users to create and edit visual content directly within the platform.
 - Added new pages and tables with full CRUD functionalities to manage product information seamlessly.